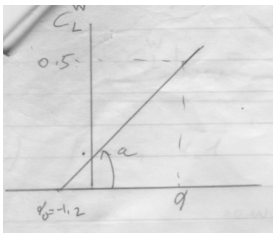
EMS612U Flight dynamics and control
Tutorial 1 typed solutions

1.1



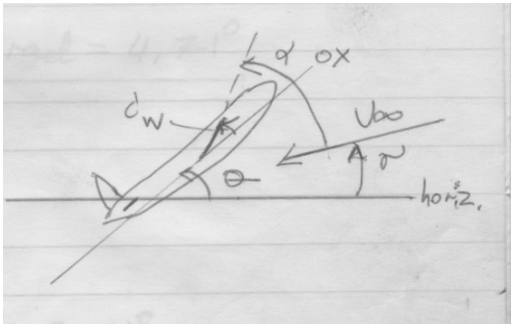
$$a=\frac{C_L^W}{\alpha-\alpha_0}$$

$$\alpha=\frac{C_L^W}{a}+\alpha_0=\frac{0.5}{4.8}-\frac{1.2\pi}{180}$$

$$\alpha=0.0832\ rad=4.77^0$$

1.2

θ = pitch angle
γ = climbing angle



Relationship:

$$\alpha+\gamma=i_w+\theta$$

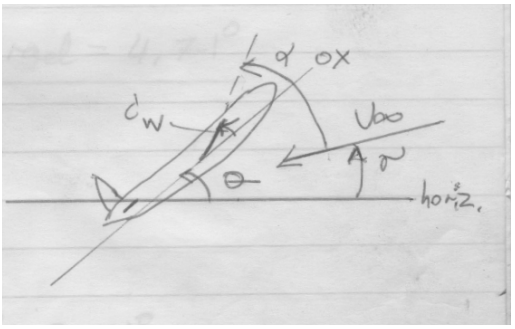
Solving for θ:

$$\theta=\alpha-i_w+\gamma=4.77^0-3^0+0^0$$

Final result:

$$\theta=1.77^0$$

1.3



Force equation normal to U_∞

$$L_T+L_W=W*\cos(\gamma)$$

$$L_T=\frac{1}{2}\rho U_\infty^2 S_T C_L^T$$

$$L_W=\frac{1}{2}\rho\cdot U_\infty^2 S_W C_L^W$$

$U_\infty, \rho, S_{T,W}$ stay the same as when γ was varied

Therefore
$C_L^W(\gamma)=C_L^W(\gamma=0)*\cos(\gamma)=0.485$

(i) From 1.1

$$\alpha = \frac{C_L^W}{a} + \alpha_0 = \frac{0.485}{4.8} - \frac{1.2\pi}{180}$$

$$\alpha = 0.082\ rad = 4.71^0$$

(ii) From 1.2

$$\theta = \alpha - i_w + \gamma = 4.71^0 - 3^0 + 8^0 = 9.71^0$$

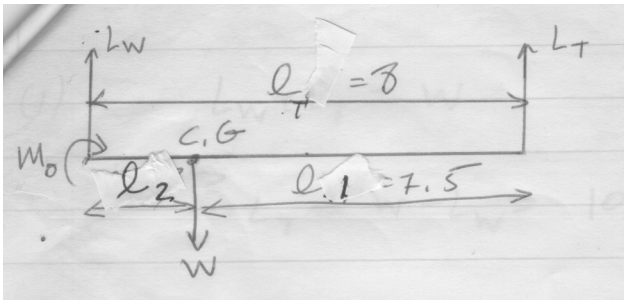
1.4

Throttle must be increased in order to overcome the contribution of the weight on the direction of U_∞ direction, i.e. $T_{new} = D + W * \sin(\gamma)$, T – thrust, D – drag

1.5

Moment equation around the tail plane

$M_0 + L_W * l_T = W * l_1$ divide the equation by $\frac{1}{2}\rho U_\infty^2 S_W \bar{\bar{c}}$



$$C_{m0} + C_L^W * (l_T / \bar{\bar{c}}) = C_W * (l_1 / \bar{\bar{c}})$$

Solving for C_W:

$$C_W = (\bar{\bar{c}} / l_1) * \left[C_{m0} + C_L^W (l_T / \bar{\bar{c}})\right]$$

$$C_W = (2.5/7.5) * (-0.037 + 0.5 * 7.5/2.5)$$

$$C_W = 0.523 \text{ for } C_L^W = 0.5$$

$$C_W = 0.518 \text{ for } C_L^W = 0.485$$

1.6

Lift coefficient definition:

$$L_W = \frac{1}{2}\rho U_\infty^2 S_W C_L^W$$

Weight coefficient definition:

$$W = \frac{1}{2}\rho U_\infty^2 S_W C_W$$

Therefore:

$$L_W / W = C_L^W / C_W$$

$$L_W = W * (C_L^W / C_W) = 10{,}000 * 9.81 * 0.5 / 0.523 = 93.7\ kN$$

1.7
(i)

$$L_W + L_T = W$$

$$L_T = W - L_W = L\,10{,}000 * 9.81 - 93.7 \times 10^3 = 4.41\,kN$$

(ii)
Divide the force equation above by $(1/2\,\rho U_\infty^2 S_W)$

$$C_L^W + C_L^T * (S_T / S_W) = C_W$$

Rearrange

$$C_L^T = (S_W / S_T) * (C_W - C_L^W)$$

$C_L^W = 0.5$ as already known, hence

$$C_L^T = (50/10) * (0.523 - 0.5) = 0.115$$

1.8

$$C_W = W / (1/2\,\rho U_\infty^2 S_W)$$

$$\Rightarrow C_{W_{new}} = C_{W_{old}} * (U_{\infty old} / U_{\infty new})^2 = 0.523/(1.2)^2 = 0.363$$

From 1.5 — Moment equation around the tail

$$C_{m0} + C_L^W * (l_T / \bar{\bar{c}}) = C_W * (l_1 / \bar{\bar{c}})$$

$$0.03 + C_L^W * (8/2.5) = 0.363 * (7.5/2.5)$$

$$C_L^W = 0.35$$

$$\alpha = \frac{C_L^W}{a} + \alpha_0 = \frac{0.35}{4.8} - \frac{1.2\,\pi}{180}$$

$$\alpha = 0.052 \text{ rad} = 2.98^0$$

Force equation:

$$C_L^T = (S_W / S_T) * (C_W - C_L^W) = (50/10) * (0.363 - 0.35) = 0.065$$

Pitch angle:

$$\theta = \alpha - i_w + \gamma = 2.98^0 - 3^0 + 0^0 = -0.02^0$$

1.9 Not solved this year (see handwritten solutions)

E Avital